

Sequences and Series

Sequence: A list of numbers demonstrating a pattern or algorithm which continues indefinitely.

Example I) $2, 4, 6, 8, \dots$ Example II) $3, 1, \frac{1}{3}, \frac{1}{9}, \dots$

Inductive Reasoning: Drawing a conclusion based on a pattern taken from limited information.

Series: A sum of numbers demonstrating a pattern or algorithm which continues indefinitely.

Example III) $5 + 25 + 125 + 625 + \dots$ Example IV) $-6 + (-3) + 0 + 3 + \dots$

Term: A number in the list of a sequence.

Find the next three terms in each sequence.

- ① $-1, -1.01, -1.02, -1.03, \dots$ ② $8, 21, 34, 47, \dots$

$-1.04, -1.05, -1.06$

$60, 73, 86$

- ③ $-1, 4, -16, 64, \dots$

$-256, 1024, -4096$

- ④ $9, 6, 4, \frac{8}{3}, \dots$

$\frac{16}{9}, \frac{32}{27}, \frac{64}{81}$

- ⑤ $10, 3, -4, -11, \dots$

$-18, -25, -32$

- ⑥ $-1, 10, -100, 1000, \dots$

$-10,000, 100,000, -1,000,000$

- ⑦ $0.75, 1, 1.25, 1.5, \dots$ ⑧ $\frac{1}{6}, \frac{1}{3}, \frac{2}{3}, \frac{4}{3}, \dots$

$1.75, 2, 2.25$

$\frac{8}{3}, \frac{16}{3}, \frac{32}{3}$

- ⑨ $1, 8, 27, 64, \dots$

$125, 216, 343$

- ⑩ $4, 9, 16, 25, \dots$

$36, 49, 64$

Arithmetic Sequence: A sequence whose consecutive terms differ by the same amount.

Example I) $2, 4, 6, 8, \dots$ Example II) $-6, -3, 0, 3, \dots$

Finding a Term in an Arithmetic Sequence

If a_n is the n^{th} term of an arithmetic sequence, then

$$a_n = a_1 + D(n-1)$$

where a_1 is the first term of the sequence, D is the common difference, and " n " is the number of the term that we are looking for (e.g. 1st term a_1 , 2nd term a_2 , 3rd term a_3 , etc.).

Find the indicated term in each arithmetic sequence below.

• (11) 9, 5, 1, ...

$$n = 82$$

$$a_n = a_1 + D(n-1)$$

$$a_{82} = 9 + (-4)(82-1)$$

$$a_{82} = 9 - 4(81)$$

$$a_{82} = 9 - 324$$

$$a_{82} = \boxed{-315}$$

(12) -50, -40, -30, ...

$$n = 23$$

$$a_n = a_1 + D(n-1)$$

$$a_{23} = -50 + 10(23-1)$$

$$a_{23} = -50 + 10(22)$$

$$a_{23} = -50 + 220$$

$$a_{23} = \boxed{170}$$

(13) 7.86, 7.88, 7.90, ...

$$n = 54$$

$$a_n = a_1 + D(n-1)$$

$$a_{54} = 7.86 + (0.02)(54-1)$$

$$= 7.86 + 0.02(53)$$

$$= 7.86 + 1.06$$

$$= \boxed{8.92}$$

(14) $\frac{3}{4}, 2\frac{3}{4}, 4\frac{3}{4}, \dots$

$$n = 16$$

$$a_n = a_1 + D(n-1)$$

$$a_{16} = \frac{3}{4} + 2(16-1)$$

$$= \frac{3}{4} + 2(15)$$

$$= \frac{3}{4} + 30$$

$$= \boxed{30\frac{3}{4}}$$

• (15) 11, 21, 31, ...

$$n = 20$$

$$a_n = a_1 + D(n-1)$$

$$a_{20} = 11 + 10(20-1)$$

$$a_{20} = 11 + 10(19)$$

$$= 11 + 190$$

$$= \boxed{201}$$

(16) -8, -2, 4, ...

$$n = 31$$

$$a_n = a_1 + D(n-1)$$

$$a_{31} = -8 + 6(31-1)$$

$$= -8 + 6(30)$$

$$= -8 + 180$$

$$= \boxed{172}$$

(17) $\frac{6}{7}, \frac{5}{7}, \frac{4}{7}, \dots$

$$n = 106$$

$$a_n = a_1 + D(n-1)$$

$$a_{106} = \frac{6}{7} + \left(-\frac{1}{7}\right)(106-1)$$

$$= \frac{6}{7} - \frac{1}{7}(105)$$

$$= \frac{6}{7} - \frac{105}{7}$$

$$= \boxed{-\frac{99}{7}}$$

(18) 92.5, 93, 93.5, ...

$$n = 87$$

$$a_n = a_1 + D(n-1)$$

$$a_{87} = 92.5 + 0.5(87-1)$$

$$= 92.5 + 0.5(86)$$

$$= 92.5 + 43$$

$$= \boxed{135.5}$$

Finding the Sum of the First "n" Terms in an Arithmetic Sequence

If " S_n " is the sum of the first "n" terms in an arithmetic sequence, then

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

where a_1 is the first term in the sequence, and D is the common difference (i.e. the number that is added to each term to find the next term).

Find the sum of the first "n" terms of each arithmetic sequence.

• (19) 1, -1, -3, ...

$$n = 11$$

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{11} = \frac{11}{2}(2(1) + (-2)(11-1))$$

$$S_{11} = \frac{11}{2}(2 - 2(10))$$

$$S_{11} = \boxed{-99}$$

(20) 11, 36, 61, ...

$$n = 37$$

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{37} = \frac{37}{2}(2(11) + 25(37-1))$$

$$= \frac{37}{2}(22 + 25(36))$$

$$= \boxed{17,057}$$

(21) $\frac{3}{2}, -\frac{1}{2}, -\frac{5}{2}, \dots$

$$n = 20$$

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{20} = \frac{20}{2}(2(\frac{3}{2}) + (-2)(20-1))$$

$$= 10(3 - 2(19))$$

$$= 10(3 - 38)$$

$$= \boxed{-350}$$

• (22) 4, 5, 6, ...

$$n = 26$$

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{26} = \frac{26}{2}(2(4) + (1)(26-1))$$

$$= 13(8 + (1)(25))$$

$$= 13(8 + 25)$$

$$= \boxed{429}$$

(23) -20, -25, -30, ...

$$n = 50$$

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{50} = \frac{50}{2}(2(-20) + (-5)(50-1))$$

$$= 25(-40 - 5(49))$$

$$= \boxed{-7,125}$$

(24) -0.115, 0.286, 0.687, ...

$$n = 10$$

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{10} = \frac{10}{2}(2(-0.115) + 0.401(10-1))$$

$$= 5(-0.23 + 0.401(9))$$

$$= \boxed{16.895}$$

Geometric Sequence: A sequence whose ratios of consecutive terms have the same value.

Example VII) 5, 25, 125, 625, ... Example VIII) $3, -1, \frac{1}{3}, -\frac{1}{9}, \dots$

Finding a Term in a Geometric Sequence

If a_n is the n th term of a geometric sequence, then

$$a_n = a_1 r^{n-1}$$

where a_1 is the first term in the sequence, " r " is the common ratio, and " n " is the number of the term that we are looking for (e.g. 1st term a_1 , 2nd term a_2 , 3rd term a_3 , etc.).

Find the indicated term in each geometric sequence below. You may leave your answers in exponential form if they are very large or very small.

• (25) $-2, -8, -32, \dots$

$$n = 7$$

$$a_n = a_1 r^{n-1}$$

$$a_7 = -2(4)^{7-1}$$

$$= -2(4)^6$$

$$= -2(4,096)$$

$$= \boxed{-8,192}$$

(26) $10,000, 1,000, 100, \dots$

$$n = 20$$

$$a_n = a_1 r^{n-1}$$

$$a_{20} = 10,000 \left(\frac{1}{10}\right)^{20-1}$$

$$= 10,000 \left(\frac{1}{10}\right)^{19}$$

$$= 10^3 \frac{1^{19}}{10^{19}}$$

$$= \boxed{\frac{1}{10^{16}}}$$

(27) $-4, 6, -9, \dots$

$$n = 32$$

$$a_n = a_1 r^{n-1}$$

$$-4r = 6$$

$$a_{32} = -4 \left(-\frac{3}{2}\right)^{32-1}$$

$$r = -\frac{3}{2}$$

$$= \boxed{-4 \left(-\frac{3}{2}\right)^{31}}$$

(28) $3, 2.4, 1.92, \dots$

$$n = 16$$

$$a_n = a_1 r^{n-1}$$

$$3r = 2.4$$

$$a_{16} = 3(0.8)^{16-1}$$

$$r = \frac{2.4}{3}$$

$$r = 0.8$$

$$= \boxed{3(0.8)^{15}}$$

• (29) $\frac{1}{2}, 1, 2, \dots$

$$n = 9$$

$$a_n = a_1 r^{n-1}$$

$$a_9 = \frac{1}{2}(2)^{9-1}$$

$$= \frac{1}{2}(2)^8$$

$$= \frac{2^8}{2}$$

$$= 2^7$$

$$= \boxed{128}$$

(30) $-4, -3, -\frac{9}{4}, \dots$

$$n = 12$$

$$a_n = a_1 r^{n-1}$$

$$-4r = -3$$

$$a_{12} = -4 \left(\frac{3}{4}\right)^{12-1}$$

$$r = \frac{3}{4}$$

$$= -4 \left(\frac{3}{4}\right)^{11}$$

$$= -4^1 \frac{3^{11}}{4^{11}}$$

$$= \boxed{-\frac{3^{11}}{4^{10}}}$$

③① $0.005, -0.015, 0.045, \dots$
 $n = 100$ $0.005r = -0.015$
 $a_n = a_1 r^{n-1}$ $r = -3$
 $a_{100} = 0.005(-3)^{100-1}$
 $= \boxed{0.005(-3)^{99}}$

③② $\frac{1}{100}, \frac{1}{25}, \frac{4}{25}, \dots$
 $n = 23$ $\frac{1}{100}r = \frac{1}{25}$
 $a_n = a_1 r^{n-1}$ $r = 4$
 $a_{23} = \frac{1}{100} 4^{23-1}$
 $= \boxed{\frac{4^{22}}{100}}$

Finding the Sum of the First "n" Terms in a Geometric Sequence

If " S_n " is the sum of the first "n" terms in a geometric sequence, then

$$S_n = a_1 \frac{1-r^n}{1-r}$$

where a_1 is the first term in the sequence, and r is the common ratio (i.e. the number that each term is multiplied by to find the next term).

Find the sum of the first "n" terms in each geometric sequence. You may leave your answers in exponential form if they are very large or very small.

• ③③ $\frac{1}{16}, \frac{3}{8}, \frac{9}{4}, \dots$
 $n = 7$

$$S_n = a_1 \frac{1-r^n}{1-r} \quad \frac{1}{16}r = \frac{3}{8}$$

$$S_7 = \left(\frac{1}{16}\right) \frac{1-6^7}{1-6} \quad r = 6$$

$$= \left(\frac{1}{16}\right) \frac{1-279,936}{-5}$$

$$= \frac{1}{16} \frac{(-279,935)}{-5}$$

$$= \frac{1}{16} (55,987)$$

$$= \boxed{3,499.1875}$$

③④ $8, -4, 2, \dots$

$$n = 18$$

$$S_n = a_1 \frac{1-r^n}{1-r}$$

$$S_{18} = 8 \frac{1-\left(-\frac{1}{2}\right)^{18}}{1-\left(-\frac{1}{2}\right)}$$

$$= 8 \frac{1-\left(\frac{1}{2}\right)^{18}}{1+\frac{1}{2}}$$

$$= 8 \frac{1-\left(\frac{1}{2}\right)^{18} \cdot 2}{\frac{3}{2} \cdot 2}$$

$$= 8 \frac{\left(1-\left(\frac{1}{2}\right)^{18}\right) \cdot 2}{3}$$

$$= \boxed{\frac{16}{3} \left(1-\left(\frac{1}{2}\right)^{18}\right)}$$

③⑤ $-0.6, -0.54, -0.486, \dots$ • ③⑥ $1, 2, 4, \dots$

$n = 50$

$-0.6r = -0.54$

$n = 7$

$S_n = a_1 \frac{1-r^n}{1-r}$ $r = 0.9$

$S_n = a_1 \frac{1-r^n}{1-r}$

$S_{50} = -0.6 \frac{1-(0.9)^{50}}{1-0.9}$

$S_7 = (1) \frac{1-2^7}{1-2}$

$= -0.6 \frac{1-(0.9)^{50}}{0.1}$

$= \frac{1-128}{-1}$

$= \boxed{-6(1-(0.9)^{50})}$

$= \frac{-127}{-1}$

$= \boxed{127}$

③⑦ $-2, -14, -98, \dots$ ③⑧ $\frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \dots$

$n = 22$

$n = 100$

$\frac{1}{3}r = \frac{1}{9}$

$S_n = a_1 \frac{1-r^n}{1-r}$

$S_n = a_1 \frac{1-r^n}{1-r}$

$r = \frac{1}{3}$

$S_{22} = -2 \frac{1-7^{22}}{1-7}$

$S_{100} = \left(\frac{1}{3}\right) \frac{1-\left(\frac{1}{3}\right)^{100}}{1-\frac{1}{3}}$

$= -2 \frac{1-7^{22}}{-6}$

$= \left(\frac{1}{3}\right) \frac{1-\left(\frac{1}{3}\right)^{100}}{\frac{2}{3}} \cdot 3$

$= \boxed{\frac{1-7^{22}}{3}}$

$= (1) \frac{1-\left(\frac{1}{3}\right)^{100}}{2}$

$= \boxed{\frac{1-\left(\frac{1}{3}\right)^{100}}{2}}$